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Landscape genetic patterns of the rainbow darter *Etheostoma caeruleum*: a catchment analysis of mitochondrial DNA sequences and nuclear microsatellites

A. E. Haponski*, T. L. Bollin*†, M. A. Jedlicka‡§ and C. A. Stepien*‖

**Great Lakes Genetics Laboratory, Lake Erie Center and Department of Environmental Sciences, University of Toledo, 6200 Bayshore Road, Toledo, OH 43616, U.S.A, †Toledo Early College High School (TECHS), The University of Toledo Scott Park Campus, Toledo, OH 43607, U.S.A. and ‡Great Lakes Genetics Laboratory at its former location, Environmental Institute, Cleveland State University, Cleveland, OH 44115, U.S.A.*

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Catchment population structure and divergence patterns of the rainbow darter *Etheostoma caeruleum* (Percidae: Teleostei), an eastern North American benthic fish, are tested using a landscape genetics approach. Allelic variation at eight nuclear DNA microsatellite loci and two mitochondrial DNA regions [cytochrome (cyt) *b* gene and control region; 2056 aligned base pairs (bp)] is analysed from 89 individuals and six sites in the Lake Erie catchment (Blanchard, Chagrin, Cuyahoga and Grand Rivers) *v*. the Ohio River catchment (Big Darby Creek and Little Miami River). Genetic and geographic patterning is assessed using phylogenetic trees, pair-wise F_{ST} analogues, AMOVA partitioning, Mantel regression, Bayesian assignment, 3D factorial correspondence and barrier analyses. Results identify 34 cyt *b* haplotypes, 22 control region haplotypes and 137 microsatellite alleles whose distributions demonstrate marked genetic divergence between populations from the Lake Erie and Ohio River catchments. *Etheostoma caeruleum* populations in the Lake Erie and Ohio River catchments diverged *c*. 1·6 mya during the Pleistocene glaciations. Greater genetic separations characterize the Ohio River populations, reflecting their older habitat age and less recent connectivity. Divergence levels within the Lake Erie catchment denote more recent post-glacial origins. Notably, the western Lake Erie Blanchard River population markedly differs from the three central basin tributary samples, which are each genetically distinguishable using microsatellites. Overall relationships among the Lake Erie sites refute a genetic isolation by geographic distance hypothesis. *Etheostoma caeruleum* populations thus exchange few genes and have low migration among tributaries and catchments.
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Key words: *Etheostoma caeruleum*; freshwater fish; Lake Erie; landscape genetics; Ohio River; Pleistocene glaciations.

INTRODUCTION

Landscape genetic methodology, which merges landscape ecology with population genetics, offers new tools for elucidating the historic and contemporary processes

‖Author to whom correspondence should be addressed: Tel.: +1 419 530 8362; fax: +1 419 530 8399; email: carol.stepien@utoledo.edu
§Present address: Cleveland Clinic, Department of Biomedical Engineering, 9500 Euclid Avenue, Cleveland, OH 44195, U.S.A.

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that shape the gene pools of native species (Manel *et al*., 2003; LeClerc *et al*., 2008). For freshwater fishes, watershed and tributary configurations are the dominant landscape features that regulate their populations. Connections and divisions among stream systems result in genetic homogenization or differentiation over time (Ward *et al*., 1994; Avise, 2000). Modern North American freshwater fish distributions were largely shaped by the Pleistocene glaciations (Costello *et al*., 2003) that occurred *c*. 1·8 million to 10 000 BP. Glacial advances isolated fish populations in southern refugia, leading to divergences and bottlenecks; and interglacial melt periods allowed them to disperse and mix (Hewitt, 1996; Lewis *et al*., 2008).

The Wisconsinan glaciations (10 000–20 000 BP) were the last of the Pleistocene epoch to affect the North American landscape, reaching their maxima *c*. 18 000 BP and isolating most of the northern fish fauna in southerly refugia. During periods of wax and wane proglacial lakes with radically different drainage patterns (Larson & Schaetzl, 2001) shaped the modern Great Lakes fish populations (Mandrak & Crossman, 1992). With the final glacial retreats, fishes dispersed and founded populations in new waterways (*c*. 4000–13 000 BP; Scott & Crossman, 1973; Bailey & Smith, 1981; Bernatchez & Wilson, 1998). Several studies found genetic evidence that Great Lakes fish populations originated from three primary refugia, Missourian, Mississippian and Atlantic Coastal (Billington *et al*., 1992; Murdoch & Hebert, 1997; Turgeon & Bernatchez, 2001; Sepulveda-Villet *et al*., 2009; Stepien *et al*., 2009). Their genetic compositions in today's catchments trace to these historic gene pools, which have since been regulated by migration or isolation through natural or anthropogenic waterway connections and habitat changes.

A landscape genetics approach is employed here to analyse the historic and contemporary genetic isolation and connectivity of fine-scale catchment patterns in the rainbow darter *Etheostoma caeruleum* Storer (Percidae: Teleostei). *Etheostoma caeruleum* is an abundant, small benthic fish that is widely distributed throughout the Mississippi and Ohio River systems, as well as all Great Lakes drainages except Lake Superior (Esmond & Stauffer, 1983). The species is monitored by the U.S. Environmental Protection Agency (EPA, 2007) as an ecological indicator, has intermediate pollution tolerance and is one of the first to disappear following habitat disturbance (Tonnis, 2006). Habitat alterations and deleterious environmental conditions probably limited historic and contemporary distributions of *E. caeruleum*, resulting in population fragmentation and genetic isolation.

Habitat specificity often limits the distribution and dispersal of darters, including *E. caeruleum* (Page, 2000). For most of the year, *E. caeruleum* inhabits gravel and rubble riffle areas in small to medium-sized streams (Page & Burr, 1991). Their greatest migrations typically occur during the spawning season when they move from deeper pools occupied during winter, and are usually limited to a few kilometres over their life history (Page, 2000). *Etheostoma caeruleum* populations, however, have expanded their range over recent decades into the Genesee, New and Potomac Rivers (Smith, 1985; Jenkins & Burkhead, 1994; Neely & George, 2006). For example, *E. caeruleum* appeared absent from the Genesee River in 1939, but by 1982 was recorded as abundant (Smith, 1985). Similarly, it was recorded as rare in the New River until the 1990s (Jenkins & Burkhead, 1994). These recent migrations may provide insight into post-glacial colonization patterns.

Etheostoma caeruleum population genetic structure is relatively unknown, and this study is the first to test its fine-scale genetic variability in the Great Lakes region

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using microsatellites. A literature search revealed only two previous genetic studies, which were based on lower resolution techniques and more limited sampling in the areas examined here. Those studies raised two different hypotheses for whether *E. caeruleum* populations in the Great Lakes and Ohio River systems are or are not differentiated, which are tested here using higher resolution genetic markers and larger sample sizes. Notably, an allozyme study by Heithaus & Laushman (1997) discerned that samples from the Lake Erie catchment (Black, Huron, Rocky and Vermilion Rivers) and the Ohio River system (sampling only the Kokosing River) are divergent. A study by Ray *et al*. (2006) based on mtDNA cytochrome (cyt) *b* sequences, however, examined very few fish (four individuals from two Lake Erie sites and 11 individuals from the Ohio River catchment) and resolved a single clade, finding no significant difference between them.

The present study analyses fine-scale population genetic structure of *E. caeruleum* among tributaries of two catchments, in relation to historic and contemporary connections, using landscape genetics methodology in a dual mitochondrial and nuclear DNA approach. Population samples from the Lake Erie catchment and Ohio River system are analysed using sequences from two mtDNA regions (cyt *b* and control region) and eight highly variable nuclear DNA microsatellite loci. Additionally, population patterns within each system and their genetic separations are evaluated.

MATERIALS AND METHODS

SAMPLING AND DNA EXTRACTION

Etheostoma caeruleum samples were collected under state collection permits using a combination of kick seining and electroshocking from six sites in the Lake Erie and Ohio River catchments, totalling 89 individuals (Table I and Fig. 1). Ohio River catchment samples include Big Darby Creek and the Little Miami River, and those from Lake Erie span its southern shoreline along the western and central basins, including the Blanchard, Chagrin, Cuyahoga and Grand Rivers (Fig. 1). Fish were anaesthetized using MS-222 under University of Toledo IACUC protocols, killed, preserved immediately in 95% ethanol and stored at room temperature. DNA was extracted from a right pectoral fin clip or *c*. 25 mg of liver tissue using a Qiagen extraction kit (www.quiagen.com), following manufacturer's directions. Extractions were assayed for quality and quantity on a 1% agarose minigel stained with ethidium bromide.

GENE AMPLIFICATION AND DNA SEQUENCING

The mtDNA cyt *b* gene and control region were amplified using the polymerase chain reaction (PCR) and protocols adapted from Stepien & Faber (1998), with the primers L-14724 and H-15915 (Schmidt & Gold, 1993) for the cyt *b* gene, and L-Pro (Porter *et al*., 2002) and 12Sar-H (Martin *et al*., 1992) for the control region. PCR was performed in 25 μl reactions containing 50 mM KCl, 1·5 mM $MgCl_2$, 10 mM Tris–HCl, 50 μM of each deoxynucleotide, 0·5 μM each of the forward and reverse primers, 30 ng DNA template and 1 unit of *Taq* polymerase. The amplification protocol was 42 cycles of 40 s at 94° C, 40 s at 52° C and 1·5 min at 72° C. A 5 μl aliquot of each PCR product was visualized on a 1% agarose minigel stained with ethidium bromide, and successful reactions were purified using a Qiagen PCR purification kit.

DNA sequencing was outsourced to the Cornell University Life Sciences Core Laboratories Center, which uses Applied Biosystems (www.appliedbiosystems.com) automated 3730 DNA analysers. For the cyt *b* gene, sequencing primers were the same as those used for PCR amplifications and the control region was sequenced using L-Pro and 12S.1-R (Porter

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TABLE I. *Etheostoma caeruleum* sampling locations, sample size (N) and summary of mitochondrial (mt) DNA and nuclear microsatellite population genetic data. For mtDNA data: number of haplotypes (N_H), number of private haplotypes (N_{PH}) and gene diversity. For microsatellite data: number of alleles (N_A), number of private alleles (N_{PA}), H_O, observed heterozygosity; H_E, expected heterozygosity; F_{IS}, inbreeding coefficient

				Cytochrome *b*				Control region			Microsatellites				
Catchment	Location	Latitude (° N)	Longitude (° W)	N	N_H	N_{PH}	Gene diversity	N_H	N_{PH}	Gene diversity	N_A	N_{PA}	H_O	H_E	F_{IS}
L. Erie	Total			58	15	15	0·533	8	8	0·443	112	21	0·813	0·844	0·116
West	Blanchard R.	41·036229	83.576722	15	9	8	0·912	3	2	0·362	75	4	0·818	0·850	0·039
East	Total			43	7	6	0·338	6	5	0·467	103	8	0·811	0·826	0·018
	Cuyahoga R.	41·159442	81.606751	15	2	1	0·133	3	1	0·448	73	3	0·742	0·748	0·016
	Chagrin R.	41·488884	81.396439	15	4	3	0·543	3	2	0·257	73	3	0·766	0·851	−0·002
	Grand R.	41·742064	81.047707	13	3	2	0·295	3	2	0·603	66	2	0·808	0·904	−0·044
Ohio R.	Total			31	19	19	0·929	14	14	0·856	116	25	0·786	0·890	0·037
	Little Miami R.	39·782865	83.876253	15	12	9	0·962	9	7	0·848	96	12	0·817	0·830	0·106
	Big Darby Ck.	39·947505	83.236055	16	10	7	0·867	7	5	0·792	84	4	0·758	0·757	0·101
Total				89	34		0·791	22		0·744	137		0·803	0·882	0·089

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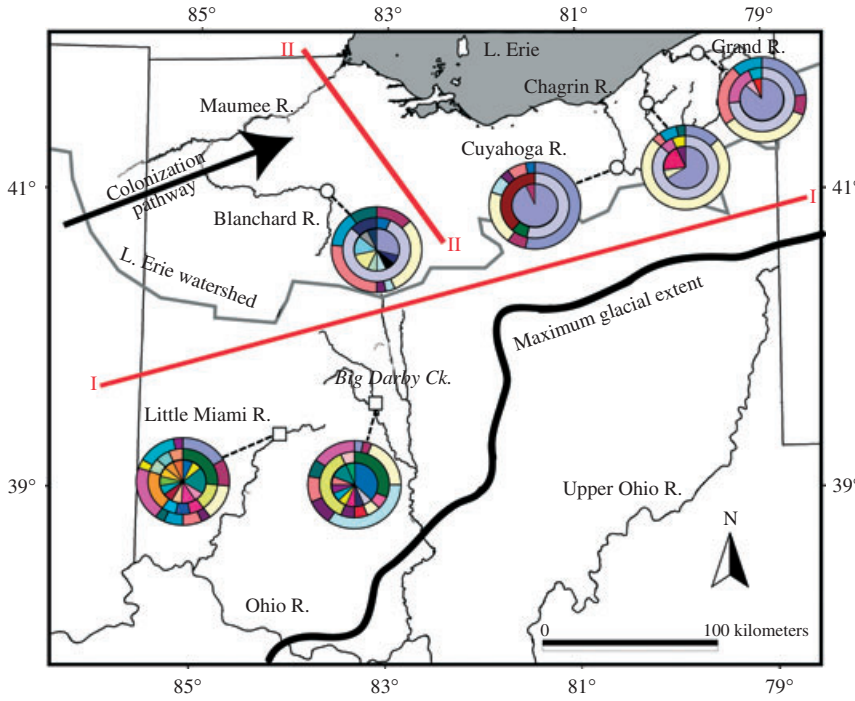

FIG. 1. Map of *Etheostoma caeruleum* sampling sites (Lake Erie group = circles, Ohio River group = squares), their primary population genetic barriers (red lines; I = strongest, according to Manni *et al.* (2004) barrier analysis) and pie charts of allelic frequencies for cytochrome *b* (inner circle), control region (middle ring) and nuclear microsatellite locus *Eca*13 (outer ring). Dark grey line = the Lake Erie watershed; black line = maximum extent of the Pleistocene glaciations; arrow = likely path from the Mississippian refugium to Lake Erie.

et al., 2002). *Etheostoma caeruleum* mtDNA sequences were checked, identified and aligned with the programme BioEdit v7.09 (Hall, 1999). Sequences were deposited in the National Institutes of Health (NIH) GenBank (www.ncbi.nlm.nih.gov), totalling 1121 bp for cyt *b* and 935 bp for the control region. In addition, *E. caeruleum* cyt *b* sequences were compared with those of Ray *et al.* (2006) from sites east of the Mississippi River and south of Lakes Erie and Ontario (totalling 47 individuals and 35 haplotypes; see Table II).

Additional variation at eight nuclear microsatellite loci designed by Tonnis (2006) is analysed to further evaluate population structure. Amplifications were 10 µl reactions of 0·6 units *Taq* polymerase, 50 µM nucleotides, 50 mM KCl, 1·5 mM $MgCl_2$, 10 mM Tris–HCl, 0·5 µM of each primer and 30 ng of template, with a sterile mineral oil overlay to maintain volume. PCR cycling parameters for a touchdown protocol followed Tonnis (2006), with addition of a final 10 min extension at 72° C. Amplification products were diluted 1:50, of which 1 µl was added to 13 µl of formamide and ABI Gene Scan 500 size standard, and then analysed in our laboratory on an ABI 3130xl Genetic Analyser with GeneMapper v3.7. Output profiles were checked to confirm allelic size variants.

DATA ANALYSES

Evolutionary relationships among the mtDNA haplotypes for each gene region were evaluated using maximum likelihood analyses in PhyML v2.4.4 (Guindon & Gascuel, 2003; http://atgc.lirmm.fr/phyml/). Support for nodes of the trees was tested using 1000 bootstrap

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TABLE II. *Etheostoma caeruleum* sequences for (a) cytochrome *b* and (b) control region including sample group, haplotype designation, number of individuals (*n*), sampling site and frequency and GenBank accession number. Unique and identical cytochrome *b* sequences from Ray *et al*. (2006) are included with identity, river site and GenBank number. *, samples from Ray *et al*. (2006). NR, haplotypes not recovered in our study

(a) Cytochrome *b*

Sample group	Haplotype	*n*	River site (*n*)	GenBank number	Ray *et al*. (2006) GenBank number
L. Erie and Wabash R.	Cb 1	42	Blanchard (5), Cuyahoga (14), Chagrin (10), Grand (11) and Sandusky (2)*	GQ250800	DQ465181–182
	Cb 2	1	Cuyahoga	GQ250801	
	Cb 3	1	Chagrin	GQ250802	
	Cb 4	3	Chagrin	GQ250803	
	Cb 5	1	Chagrin	GQ250804	
	Cb 6	1	Grand	GQ250805	
	Cb 16	1	Blanchard	GQ250815	
	Cb 17	1	Blanchard	GQ250816	
	Cb 18	1	Blanchard	GQ250817	
	Cb 19	1	Blanchard	GQ250818	
	Cb 20	2	Blanchard	GQ250819	
	Cb 21	2	Blanchard	GQ250820	
	Cb 31	1	Grand	GQ250830	
	Cb 32	1	Blanchard	GQ250831	
	Cb 33	1	Blanchard	GQ250832	
	100	1	Tippecanoe*	NR	DQ465100
	107	1	Tippecanoe*	NR	DQ465107
	122	2	Sugar Ck.*	NR	DQ465104 & 122
	164	2	Twentymile Ck.*	NR	DQ465163–164
Green R.	078	1	Green*	NR	DQ465078
	082	2	Green*	NR	DQ465082 & 226
	197	1	Green*	NR	DQ465197
	199	2	Green*	NR	DQ465198–199
Ohio R.	Cb 7	9	Little Miami (1) and Big Darby Ck. (8)	GQ250806	DQ465110
	Cb 8	1	Big Darby Ck.	GQ250807	
	Cb 9	1	Big Darby Ck.	GQ250808	
	Cb 10	1	Big Darby Ck.	GQ250809	
	Cb 11	2	Little Miami (1) and Big Darby Ck. (1)	GQ250810	
	Cb 12	1	Big Darby Ck.	GQ250811	
	Cb 13	1	Big Darby Ck.	GQ250812	

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TABLE II. Continued

(a) Cytochrome *b*

Sample group	Haplotype	*n*	River site (*n*)	GenBank number	Ray *et al.* (2006) GenBank number
	Cb 14	1	Big Darby Ck.	GQ250813	
	Cb 15	9	Little Miami (5), Big Darby Ck. (2), Tygarts Ck. (1)* and Goose Ck. (1)*	GQ250814	DQ465142, 167 & 178
	Cb 22	1	Little Miami	GQ250821	
	Cb 23	2	Little Miami	GQ250822	
	Cb 24	1	Little Miami	GQ250823	
	Cb 25	1	Little Miami	GQ250824	
	Cb 26	2	Little Miami	GQ250825	DQ465177
	Cb 27	1	Little Miami	GQ250826	
	Cb 28	1	Little Miami	GQ250827	
	Cb 29	1	Little Miami	GQ250828	
	Cb 30	1	Little Miami	GQ250829	
	Cb 34	1	Big Darby Ck.	GQ250833	
	074	1	Tennessee*	NR	DQ465074
	077	1	Kentucky*	NR	DQ465077
	079	1	Tennessee*	NR	DQ465079
	080	1	Cumberland*	NR	DQ465080
	081	1	Cumberland*	NR	DQ465081
	102	1	Big Darby Ck.	NR	DQ465102
	105	1	Salt*	NR	DQ465105
	143	1	Kentucky*	NR	DQ465143
	152	1	Guyandotte*	NR	DQ465152
	153	1	Guyandotte*	NR	DQ465153
	202	1	Tennessee*	NR	DQ465202
	204	1	Tennessee*	NR	DQ465204
	205	1	Highland Ck.*	NR	DQ465205
Upper Ohio R. and L. Ontario	125	1	Kanawha*	NR	DQ465125
	137	2	Kanawha*	NR	DQ465137 & 193
	146	3	Genesee*	NR	DQ465144–146
	147	1	Genesee*	NR	DQ465147
	162	1	Allegheny*	NR	DQ465162
	179	1	Kokosing*	NR	DQ465179
	180	1	Kokosing*	NR	DQ465180
	186	3	Middle Fork Little Beaver* (2) and Allegheny*	NR	DQ465185–186 & 188
	187	1	Allegheny*	NR	DQ465187
	196	1	Kanawha*	NR	DQ465196

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TABLE II. Continued

(b) Control region

Group	Haplotype	*n*	River site (*n*)	GenBank number
L. Erie	Cr 7	1	Blanchard	GQ250840
	Cr 8	43	Blanchard (12), Cuyahoga (11), Chagrin (13) and Grand (7)	GQ250841
	Cr 9	2	Blanchard	GQ250842
	Cr 10	4	Cuyahoga (3) and Chagrin (1)	GQ250843
	Cr 11	1	Chagrin	GQ250844
	Cr 12	1	Cuyahoga	GQ250845
	Cr 13	1	Grand	GQ250846
	Cr 14	5	Grand	GQ250847
Ohio R.	Cr 1	11	Little Miami (6) and Big Darby Ck. (5)	GQ250834
	Cr 2	1	Big Darby Ck.	GQ250835
	Cr 3	1	Big Darby Ck.	GQ250836
	Cr 4	1	Big Darby Ck.	GQ250837
	Cr 5	1	Big Darby Ck.	GQ250838
	Cr 6	8	Little Miami (2) and Big Darby Ck. (6)	GQ250839
	Cr 15	1	Little Miami	GQ250848
	Cr 16	1	Little Miami	GQ250849
	Cr 17	1	Little Miami	GQ250850
	Cr 18	1	Little Miami	GQ250851
	Cr 19	1	Little Miami	GQ250852
	Cr 20	1	Little Miami	GQ250853
	Cr 21	1	Little Miami	GQ250854
	Cr 22	1	Big Darby Ck.	GQ250855

Note: Ray *et al*. (2006) assigned GenBank numbers to all individuals sampled, whereas this paper's home laboratory policy is to assign numbers to unique haplotypes.

pseudo-replications (Felsenstein, 1985). Akaike information criterion (AIC) from Modeltest v3.7 (Posada & Crandall, 1998; http://darwin.uvigo.es/software/modeltest.html) was used to determine the most appropriate model of substitution for the cyt *b* and control region datasets for the maximum likelihood analyses. The Tamura & Nei (1993) model including invariable sites (0·530) and a gamma distribution ($\alpha = 1{\cdot}358$) was selected for the cyt *b* dataset, and the Hasegawa, Kishino and Yano (HKY) model (Hasegawa *et al*., 1985) with a gamma distribution ($\alpha = 0{\cdot}185$) was designated as the most appropriate model for the control region sequence data. The programme TCS v1.21 (Clement *et al*., 2000) was used to construct a statistical parsimony cyt *b* haplotype network to further analyse relationships and facilitate comparisons with the study by Ray *et al*. (2006).

A partition homogeneity test was conducted to compare the similarity of phylogenetic signal among sequence data from the mtDNA cyt *b* gene and control region to evaluate whether the datasets could be reliably combined in a total evidence approach (Farris *et al*.,

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1994, 1995). The test showed that the two datasets were incongruent. Uninformative sites were removed and analyses re-run; the datasets, however, remained incongruent and thus were not combined.

Divergence times among *E. caeruleum* cyt *b* lineages were evaluated using a penalized likelihood approach (Sanderson, 2002) implemented in the programme r8s v1.71 (Sanderson, 2003). Initially, the complete cyt *b* dataset was tested for conformance to a molecular clock model, from which it significantly varied ($P = 0{\cdot}01$). A second analysis using penalized likelihood was conducted with an optimal smoothing parameter ($= 3{\cdot}00$) determined using cross-validation in r8s. Divergence time estimates under this method require fossil calibration points for at least one node. Because darters are very poorly preserved in the fossil record (Cavender, 1986), fossil dates and sequences from other members of the family Percidae were included to calibrate the phylogeny. Cytochrome *b* sequences found in GenBank representing species from the remaining two subfamilies (Luciopercinae and Percinae) of Percidae were added, including *Gymnocephalus* (AF386598, Near, 2002; AF546114, G. R. Moyer, B. L. Sloss, C. Krajewski & N. Billington, unpubl. data; AY374279, Sloss *et al*., 2004), *Perca* (AF546115, AF546117, AF546120, Moyer *et al*., unpubl. data), *Romanichthys* (AF546121, Moyer *et al*., unpubl. data), *Sander* (AF386602, Near, 2002; AF546122, Moyer *et al*., unpubl. data; AY374290, AY374292, Sloss *et al*., 2004) and *Zingel* (AF546123, AF546124, Moyer *et al*., unpubl. data). Fossil dates used in the analyses for origins of three Percidae genera included the early Pleistocene epoch (*c*. 1·8 mya) for *Gymnocephalus* (Holcik & Hensel, 1974), Miocene (*c*. 26 mya) for *Perca* (Lebedev, 1952) and Pleistocene epoch (*c*. 1·8 mya) for *Sander* (Cavender, 1986). Thus, ages of the nodes for each of the three outgroup genera were set to those fossil dates, determined from the literature.

Population genetic data statistics for the mtDNA datasets were calculated using Arlequin v3.11 (Excoffier *et al*., 2005, 2007), including measures of haplotype diversity, genetic divergence and pair-wise comparisons. The latter were evaluated using unbiased θ estimates of F-statistics (Weir & Cockerham, 1984) and non-parametric exact tests of differentiation (χ^2 contingency table analyses; Raymond & Rousset, 1995*a*). Use of F_{ST} analogues in the present study facilitated comparisons with other studies and provided levels of comparative genetic divergence. F-statistic estimates, however, assume a normally distributed dataset (Weir & Cockerham, 1984) and may be affected by small sample sizes and rare alleles (Raymond & Rousset, 1995*a*). Thus, pair-wise exact tests of differentiation also were run that used a Markov-Chain Monte-Carlo (MCMC) procedure to randomly sample allele frequencies, as they are unaffected by sample size and are not dependent on normal distribution (Raymond & Rousset, 1995*a*). Probability values for both types of multiple pair-wise comparisons were adjusted using the sequential Bonferroni method (Rice, 1989). Results of the Bonferroni correction were compared with the false discovery rate method of Benjamini & Hochberg (1995). The two methods were found to be congruent, thus only the Bonferroni value is presented here. These tests similarly were employed to evaluate pair-wise relationships using microsatellite data with the programmes *F*stat v2.9.3.2 (Goudet, 1995) for unbiased θ_{ST} (Weir & Cockerham, 1984) and Genepop v4.0 (Raymond & Rousset, 1995*b*; Rousset, 2007) for exact tests of differentiation (Raymond & Rousset, 1995*a*, *b*).

Genetic relationships among population groups were tested using several approaches. Hierarchical partitioning of variation between the two catchment groups and among their sampling locations was tested using analysis of molecular variation (AMOVA; Excoffier *et al*., 1992), separately for both mtDNA datasets and the microsatellite dataset. The hypothesis of geographic isolation (measured as nearest connected waterway paths among sampling sites) with genetic distance (unbiased θ_{ST}) was evaluated separately for all three datasets using a Mantel (1967) test, 1000 rearrangements and Isolde in Genepop. In addition, mismatch distribution analyses (Rogers & Harpending, 1992) were calculated in Arlequin using the complete cyt *b* dataset to test the null model of rapid population expansion of *E. caeruleum* populations into formerly glaciated regions. Chi-square tests were conducted to test the goodness of fit of the dataset to this null model.

Microsatellite loci were tested for conformance to Hardy–Weinberg (H-W) equilibrium expectations and linkage disequilibrium for each population sample using Genepop, and levels of significance were adjusted using Bonferroni correction (Zar, 1999). Micro-Checker v2.2.3 (van Oosterhout *et al*., 2004) evaluated each locus for the presence of null alleles,

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scoring errors or large allele dropout. For each locus, average observed heterozygosity (H_O), inbreeding coefficient (F_{IS}) and divergence (F_{ST}) were calculated in the programme *F*stat.

Genetic comparisons among the six sampling sites using microsatellite data included: observed (H_O) and expected (H_E) heterozygosities in Genepop, the inbreeding coefficient (F_{IS}) in *F*stat and the analyses used for mtDNA data that are presented above. Three-dimensional factorial correspondence analysis (3D-FCA; Benzecri, 1973) in the programme Genetix v4.05 (Belkhir *et al*., 2004), which makes no *a priori* assumptions, further explored relationships among and within populations. Relationships among locations were analysed with neighbour-joining trees (Saitou & Nei, 1987) and Cavalli-Sforza & Edwards (1967) chord distances in Phylip v3.68 (Felsenstein, 2005). Support values for nodes were evaluated using 1000 bootstrap pseudo-replications (Felsenstein, 1985) in Phylip.

The nuclear microsatellite data also were used in the Bayesian-based clustering programme Structure v2.2 (Pritchard *et al*., 2000) to test assignment of individuals to population groups of $K = 1$ (the null hypothesis of panmixia) to $K = 6$ (total number of sampling sites). Structure was run with 10 replicates for each K, a burn-in of 100 000 generations and run lengths of 2 000 000 generations. The $\Delta K{:}K$ ratio method (Evanno *et al*., 2005) and posterior probabilities (Pritchard *et al*., 2000) from each run were then used to evaluate the best K, for example, the number of true population groups in the samples.

The landscape genetics approach of Monmonier's (1973) maximum difference algorithm in Barrier v2.2 (Manni *et al*., 2004) was tested for significant barriers to gene flow using the microsatellite dataset. Barrier identifies borders among neighbouring sample site locations, by comparing the relationship between geographical coordinates (latitude and longitude) and θ_{ST} genetic distances (Manni *et al*., 2004). Barriers are designated when genetic breaks occur, characterized by high divergence despite geographic proximity. Barrier analyses followed Strange & Stepien (2007) and LeClerc *et al*. (2008) and were run separately using the overall microsatellite dataset (all eight loci) and each individual locus (the latter to determine the number of loci supporting each barrier).

RESULTS

OVERALL GENETIC PATTERNS AMONG *ETHEOSTOMA CAERULEUM* SAMPLES

Sequence data total 2056 aligned bp, with 34 mtDNA cyt *b* (GenBank accession numbers GQ250800–GQ250833) and 22 control region haplotypes (GQ250834–GQ250855), and 137 alleles from eight microsatellite loci recovered from 89 rainbow darter individuals (Table I). A marked genetic divergence of 11 nucleotide changes separates the group of 32 mtDNA cyt *b* haplotypes from the Ohio River catchment *v*. the group of 19 haplotypes from the Lake Erie catchment and the Wabash River. These two primary catchment population groups share no haplotypes (Figs 1–4) and respectively house 25 and 21 private microsatellite alleles (Table I). MtDNA and microsatellite allelic differences among the sampling sites are illustrated in Fig. 1.

This study recovers one haplotype from the Lake Erie catchment and three from the Ohio River that were described by Ray *et al*. (2006; Table II). The present study identifies 30 additional haplotypes, 14 from the Lake Erie catchment and 16 from the Ohio River system, that were not found by the Ray *et al*. (2006) study; and uses data from both investigations in a comprehensive re-analysis. Ray *et al*. (2006) described one additional Lake Erie catchment haplotype and 13 additional Ohio River haplotypes that were not found in the present study. They also sampled tributaries of three other sites for only one to two individuals, including the Wabash River (which has three unique haplotypes that are closely related to Lake Erie types), the Green

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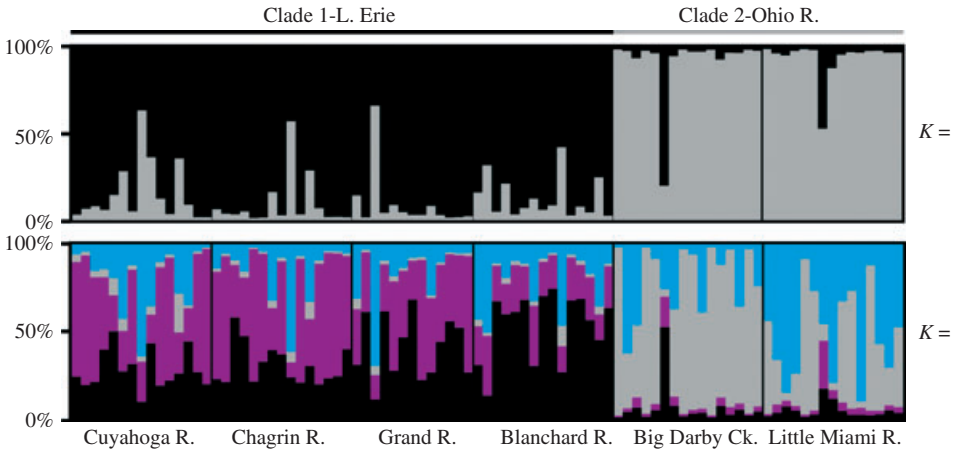

FIG. 2. Bayesian Structure v2.2 (Pritchard *et al.*, 2000) assignments for $K = 2$ and 4 *Etheostoma caeruleum* population groups based on nuclear microsatellite data. Individuals are represented by thin vertical lines and black lines separate the six sampling locations.

River in Kentucky (which appears related to the Lake Erie–Wabash River system and has four distinctive haplotypes) and the Lake Ontario–Upper Ohio River re-analysed here (which form a cluster of 10 indigenous haplotypes; see Table II and Fig. 4).

All microsatellite loci are unlinked and conform to H-W equilibrium expectations following Bonferroni correction, except locus *Eca*11 from the Blanchard River ($P = 0{\cdot}0001$), which is attributable to chance alone and thus is included in the analyses. There are some slight heterozygote deficiencies, which are not significant, as well as possible indication of null alleles at loci *Eca*48 in the Blanchard River, *Eca*11 and *Eca*49 from Big Darby Creek and *Eca*14 and *Eca*49 from Little Miami River. Because these loci are in H-W equilibrium, they are included in analyses. Repeat number and size, number of alleles, observed heterozygosities, average inbreeding coefficient and average genetic divergence for each locus are reported in Table III.

Gene diversity values total 0·791 (range = 0·133–0·962) for the cyt *b* gene, 0·744 (0·257–0·848) for the control region and 0·803 (0·742–0·818) for the microsatellite data. Heterozygosities for microsatellite loci average 0·805 (range = 0·718–0·880), with 11–23 alleles per locus (Table III). Average genetic divergences per locus range from 0·017 to 0·130, with *Eca*11 and *Eca*13 being the most informative (Table III).

DIVISION BETWEEN LAKE ERIE AND OHIO RIVER CATCHMENTS

Rainbow darters from the Lake Erie and Ohio River catchments comprise two distinct population groups, which share no mtDNA haplotypes (Fig. 1) and respectively have 25 and 21 unique microsatellite alleles (Table I). The mt and nuclear DNA datasets reveal long-term marked divergence between them, with high support in the Bayesian Structure analysis (0·99; Fig. 2) and respective monophyly in the neighbour-joining tree (79 and 88% nodal support; Fig. 3). Pair-wise comparisons between the catchment groups using θ_{ST} and exact tests of differentiation also show high divergence ($\theta_{ST} = 0{\cdot}165$–$0{\cdot}913$ for cyt *b*, 0·106–0·591 for control region and

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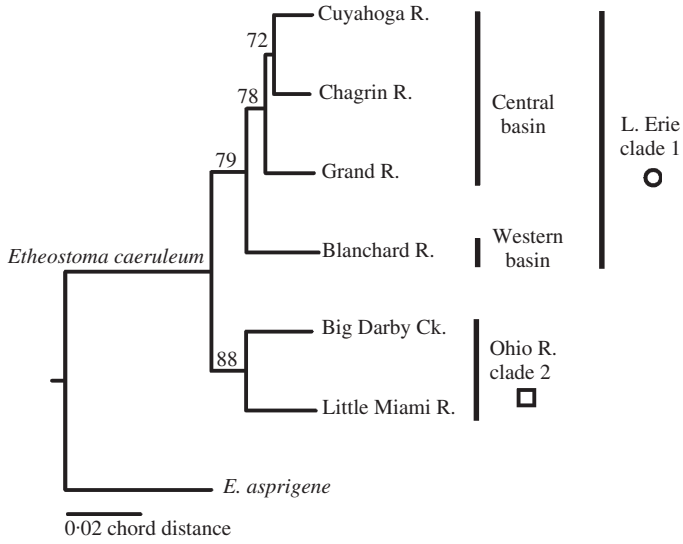

FIG. 3. Neighbour-joining tree of *Etheostoma caeruleum* population relationships based on Cavalli-Sforza & Edwards (1967) chord distances and nuclear microsatellite data, with per cent (%) bootstrap support.

0·046–0·118 for microsatellites; Table IV), as well as barrier (Fig. 1) and 3D-FCA analyses (Fig. 5). The cyt *b* haplotype maximum likelihood tree resolves two primary clades from the Lake Erie–Wabash River (91%; labelled clade 1) and Ohio River (labelled clade 2), which respectively are distinguished by 19 and 32 unique haplotypes (Table I and Fig. 4). The Lake Erie group has lower gene diversity values than does the Ohio River group, with the latter housing more mtDNA haplotypes and more microsatellite alleles (Table I and Figs 1 and 4). Additional population structure within the two catchment groups is recovered in the Structure analysis, showing $K = 4$ population groups and further subdivision among sites (Fig. 2).

The maximum likelihood tree [Fig. 4(a)] recovers four divergent *E. caeruleum* population groups using data from the present study supplemented by those from Ray *et al*. (2006), comprising the Lake Erie–Wabash River (clade 1a), Green River of Kentucky (1b), Ohio River (2a) and Lake Ontario–Upper Ohio River (2b) catchments. These cluster into two primary *E. caeruleum* population groups: (1) the Lake Erie–Wabash–Green Rivers and (2) the Ohio River–Lake Ontario–upper Ohio River, which diverged *c*. 1·6 mya based on fossil calibrations. *Etheostoma caeruleum* from the Lake Erie–Wabash River (1a) and the Green River (1b) appear as sister groups that diverged *c*. 1·09 mya. *Etheostoma caeruleum* from the main Ohio River system (2a) form the sister group of those inhabiting the Lake Ontario–upper Ohio River region (2b), diverging *c*. 1·12 mya.

POPULATION GENETIC PATTERNS WITHIN THE LAKE ERIE CATCHMENT

West to east genetic patterns are evident across the Lake Erie populations of *E. caeruleum*. All Lake Erie samples share cyt *b* haplotype Cb 1 and control region haplotype Cr 8, both of which increase in proportional frequency to the east (Fig. 1).

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TABLE III. *Etheostoma caeruleum* microsatellite loci amplified using primers designed by Tonnis (2006), with our data for repeat numbers (R_N) and sizes (R_S; bp), number of alleles (N_A), first, second and third most common alleles and their frequency of occurrence (%) for the sampled individuals, mean observed heterozygosity (H_O), average inbreeding coefficient (F_{IS}) and mean genetic divergence (F_{ST}) per locus

Locus	R_N	R_S (bp)	N_A	First (%)	Second (%)	Third (%)	H_O	F_{IS}	F_{ST}
*Eca*10	8–21	200–252	14	212 (21·3)	216 (14·6)	228 (12·9)	0·876	−0·042	0·058
*Eca*11	17–42	194–294	23	222 (16·9)	202 (15·7)	274 (12·4)	0·725	0·120	0·117
*Eca*13	6–18	164–212	13	180 (31·5)	172 (18·0)	188 (12·4)	0·775	−0·043	0·130
*Eca*14	8–18	164–204	11	180 (42·1)	196 (10·1)	176 (8·4)	0·718	0·030	0·067
*Eca*46	6–30	196–292	22	244 (10·1)	240 (9·0)	256 (8·4)	0·879	0·057	0·017
*Eca*48	15–29	170–226	16	210 (15·2)	202 (15·2)	206 (12·9)	0·809	0·079	0·033
*Eca*49	11–30	160–236	19	204 (13·5)	200 (13·5)	212 (11·2)	0·779	0·132	0·029
*Eca*71	9–30	168–252	19	192 (27·5)	196 (11·2)	200 (9·6)	0·880	−0·025	0·034

Cytochrome *b* haplotype diversity values decrease from west to east, with the western Blanchard River population having higher values (0·912) than those characterizing the central basin tributaries (0·338; Table I). Observed microsatellite heterozygosity of *E. caeruleum* samples from the four Lake Erie sites averages 0·813, ranging from 0·742 to 0·818 (Table I). There is a decrease in number of alleles from west to east across Lake Erie at locus *Eca*13 (Fig. 1), similar to the pattern observed in the mtDNA data.

Pair-wise comparisons using microsatellite data support significant divergence among all Lake Erie populations sampled [Table IV(c)], also indicated by AMOVA analyses ($\Phi_{SC} = 0{\cdot}079$). Exact tests of genetic differentiation between pairs of Lake Erie populations using mtDNA cyt *b* data indicate that the Blanchard River (western) location significantly differs from the three central basin tributaries [Table IV(a)]. The eastern-most location from the Grand River significantly differs from the other Lake Erie samples before Bonferroni correction using the control region data [Table IV(b)]. F_{ST} analogue tests reveal some divergences between the central Lake Erie basin tributary samples; with the control region data differentiating between the Grand and

FIG. 4. (a) Maximum likelihood tree of *Etheostoma caeruleum* mitochondrial cytochrome *b* haplotypes, with haplotype number and identity in parentheses for major rivers with % bootstrap support. *, haplotypes from Ray *et al*. (2006; see Table 2). Outgroup genera from the subfamilies Luciopercinae and Percinae including *Gymnocephalus* (AF386598, Near, 2002; AF546114, G. R. Moyer, B. L. Sloss, C. Krajewski & N. Billington, unpubl. data; AY374279, Sloss *et al*., 2004) *Perca* (AF546115, AF546117, AF546120, G. R. Moyer, B. L. Sloss, C. Krajewski & N. Billington, unpubl. data), *Romanichthys* (AF546121, G. R. Moyer, B. L. Sloss, C. Krajewski & N. Billington, unpubl. data), *Sander* (AF386602, Near, 2002; AF546122, G. R. Moyer, B. L. Sloss, C. Krajewski & N. Billington, unpubl. data; AY374290, AY374292, Sloss *et al*., 2004) and *Zingel* (AF546123, AF546124, G. R. Moyer, B. L. Sloss, C. Krajewski & N. Billington, unpubl. data) are included for fossil calibration points to date divergence times of rainbow darter lineages (see Section Results). (b) Parsimony network among cytochrome *b* haplotypes from TCS v1.21 (Clement *et al*., 2000). Circle sizes are proportional to number of individuals per haplotype, small empty circles = hypothetical intermediate haplotype, *, haplotypes from Ray *et al*. (2006). Haplotypes are labelled by site and thin black lines represent those that are shared among sites (see Table II).

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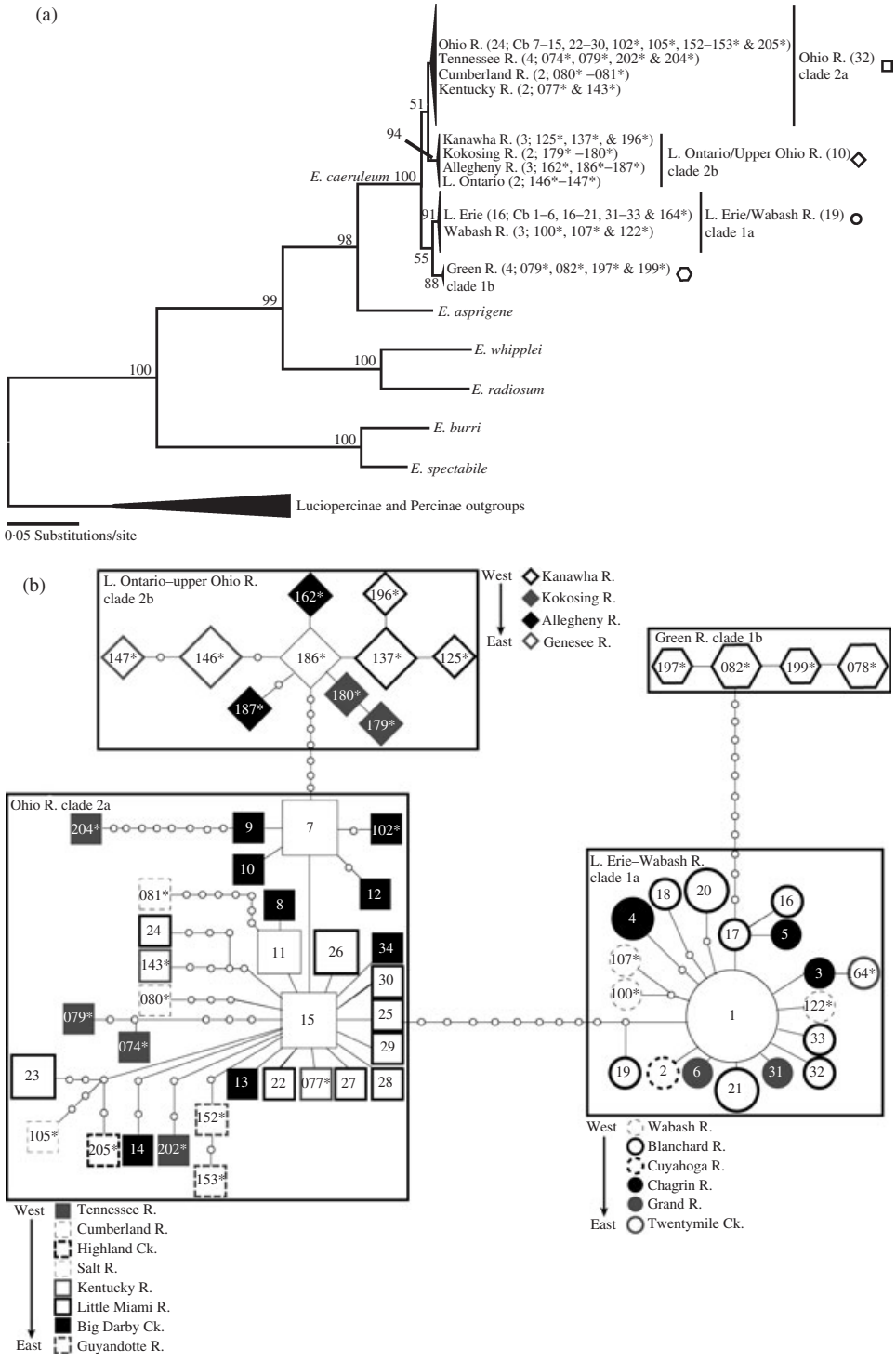

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TABLE IV. Pair-wise comparisons between *Etheostoma caeruleum* sampling sites using θ_{ST} (below the diagonal; Weir & Cockerham, 1984) and exact tests of differentiation (above the diagonal; Raymond & Rousset, 1995*a*) for (a) cytochrome *b*, (b) control region and (c) nuclear microsatellites

(a) Cytochrome *b*

	L. Erie catchment				Ohio R. catchment	
Site	Blanchard R.	Cuyahoga R.	Chagrin R.	Grand R.	Little Miami R.	Big Darby Ck.
Blanchard R.	—	**	*	*	**	**
Cuyahoga R.	0·001	—	NS	NS	**	**
Chagrin R.	0·001	0·107	—	NS	**	**
Grand R.	0·000	0·004	0·087	—	**	**
Little Miami R	0·165**	0·886**	0·859**	0·875**	—	*
Big Darby Ck.	0·186**	0·913**	0·886**	0·903**	0·118**	—

(b) Control region

	L. Erie catchment				Ohio R. catchment	
Site	Blanchard R.	Cuyahoga R.	Chagrin R.	Grand R.	Little Miami R.	Big Darby Ck.
Blanchard R.	—	NS	NS	*	**	**
Cuyahoga R.	0·161	—	NS	*	**	**
Chagrin R.	0·000	0·038	—	*	**	**
Grand R.	0·192**	0·145*	0·000	—	**	**
Little Miami R	0·288**	0·106*	0·196*	0·268**	—	NS
Big Darby Ck.	0·591**	0·349**	0·505**	0·571**	0·098	—

(c) Nuclear microsatellites

	L. Erie catchment				Ohio R. catchment	
Site	Blanchard R.	Cuyahoga R.	Chagrin R.	Grand R.	Little Miami R.	Big Darby Ck.
Blanchard R.	—	**	**	**	**	**
Cuyahoga R.	0·046**	—	**	**	**	**
Chagrin R.	0·063**	0·049**	—	**	**	**
Grand R.	0·029**	0·031**	0·033**	—	**	**
Little Miami R	0·055**	0·046**	0·088**	0·052**	—	**
Big Darby Ck.	0·074**	0·080**	0·118**	0·088**	0·027**	—

NS, not significant.
*, significant at $P = 0{\cdot}05$; **, significant after sequential Bonferroni correction (Rice, 1989).

Cuyahoga River samples before sequential Bonferroni correction, and the Grand and Blanchard River samples significantly differing both before and after correction [Table IV(b)].

Barrier analysis using the microsatellite data divides the western Lake Erie population (Blanchard River) from the central basin sites (Chagrin, Cuyahoga and Grand Rivers), and differentiates all four as separate populations. Similarly, in the Structure analysis, a peak is discerned for $K = 4$ population groups using the $\Delta K{:}K$ ratio method, separating the Blanchard River individuals from the central Lake Erie basin tributaries (Fig. 2). The chord distance neighbour-joining tree (Fig. 3) and 3D-FCA results (Fig. 5) also reveal this pronounced division. This pattern does not fit a genetic isolation by geographic distance hypothesis, according to Mantel (1967) tests for

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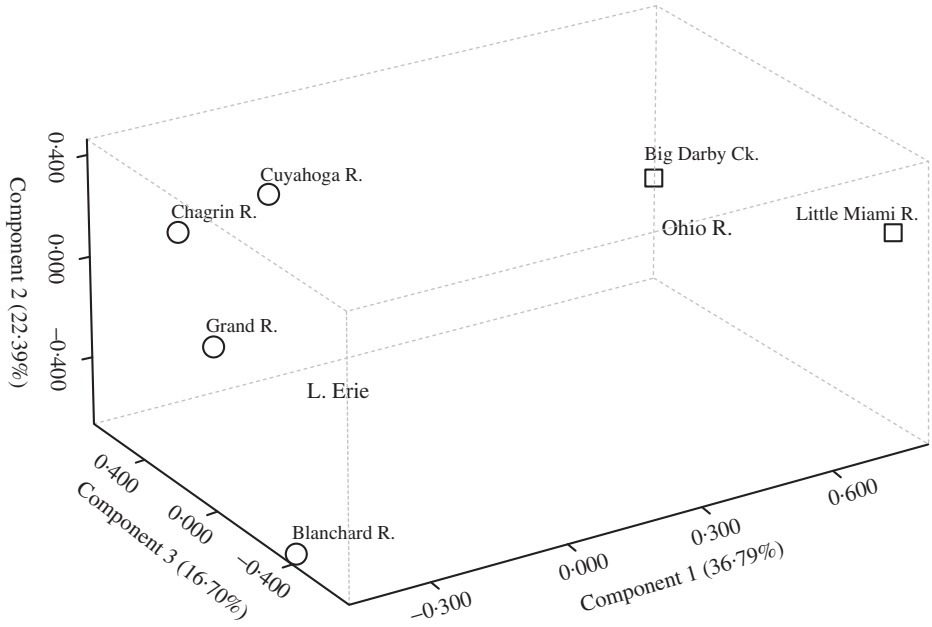

FIG. 5. Relationships among *Etheostoma caeruleum* samples using 3D factorial correspondence analysis (Benzecri, 1973) for nuclear microsatellite data. Clusters correspond to Lake Erie (circles) and Ohio River (squares) catchment locations, with those from Blanchard River also distinct from other sites.

cyt *b* and the microsatellite data, but has borderline support using the control region dataset alone ($P = 0{\cdot}04$; $y = 0{\cdot}0465x - 0{\cdot}1091$, $r^2 = 0{\cdot}2124$; figure not shown).

The cyt *b* haplotype network [Fig. 4(b)] shows that *E. caeruleum* haplotypes from the Wabash River sequenced by Ray *et al*. (2006) are one to two mutational steps from the most common Lake Erie haplotype (designated Cb 1). All other cyt *b* haplotypes in Lake Erie similarly differ by one to two mutational steps, thus the Wabash River population appears closely related to Lake Erie populations. In addition, the maximum likelihood tree [Fig. 4(a)] resolves a clade in the Green River of Kentucky to the south-west of Lake Erie (88% support), from analysis including the Ray *et al*. (2006) data. Samples from Lake Erie–Wabash River system appear related to the Green River sample in the haplotype network analysis [clades 1a and 1b; Fig. 4(b)], diverging by 13 mutational steps and *c*. 1·09 mya. Both the Lake Erie–Wabash River and Green River population groups show evidence of past rapid population expansion into the formerly glaciated regions, supported by the mismatch distribution analysis and the star pattern of the network analysis [Fig. 4(b)].

POPULATION GENETIC PATTERNS WITHIN THE OHIO RIVER CATCHMENT

Samples from Big Darby Creek and the Little Miami River share cyt *b* haplotypes Cb 7 and 15 and control region haplotypes Cr 1 and Cr 6 (Table II and Fig. 1), which increase in frequency from west to east, similar to the pattern of Lake Erie haplotypes (Fig. 1). The Ohio River sampling sites have more haplotypes and higher haplotypic diversities than those in Lake Erie, but are similar in microsatellite

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diversity (Table I). Diversity values for the Ohio River group decrease from west to east, with the Little Miami River sample having more haplotypes than does Big Darby Creek (Table I and Fig. 5). Number of microsatellite alleles also decreases from west to east at locus *Eca*13 (Fig. 1). For the microsatellite data, observed heterozygosities for the Ohio River samples total 0·786 (range = 0·758–0·817) and are very similar to the expected heterozygosities, as well as to values in the Lake Erie catchment (Table I). Pair-wise θ_{ST} comparisons and exact tests of differentiation reveal significant divergence between populations from Big Darby Creek and Little Miami River based on cyt *b* sequence [$\theta_{ST} = 0{\cdot}094$; Table IV(a)] and microsatellite data [$\theta_{ST} = 0{\cdot}027$; Table IV(c)]. Structure, 3D-FCA and AMOVA results confirm this marked population divergence (Figs 2 and 5).

Cytochrome *b* haplotypes in the Ohio River catchment reflect more genetic structure than is found in the Lake Erie–Wabash River clade, having more haplotypes and greater overall divergence [Fig. 4(b)]. There are two common haplotypes (Cb 7 and Cb 15) in the Ohio River system, which are each related to other less-common haplotypes that vary by one to six mutational steps [Fig. 4(b)]. In addition to unique haplotypes that characterize Little Miami River and Big Darby Creek samples, Ray *et al*. (2006) described related haplotypes (Table II) that uniquely distinguish samples from Highland Creek (IN) and the Cumberland (KY, TN), Guyandotte (WV), Kentucky (KY), Salt (KY) and Tennessee Rivers (AL, TN); all of which cluster with the Ohio River haplotypes (clade 2a; Fig. 4). Several cyt *b* haplotypes in these south-western locations are one to 11 mutational steps from the most common Ohio River types, Cb 7 and Cb 15, which are greater than the divergences characterizing those found in Little Miami River and Big Darby Creek [Fig. 4(b); clade 2a]. The Lake Ontario–Upper Ohio River sites from Ray *et al*. (2006) form a distinct group in the maximum likelihood tree with 94% support [clade 2b; Fig. 4(a)], which is unrelated to types from Lake Erie [$\theta_{ST} = 0{\cdot}4$; clade 1a; Fig. 4(b)]. The Ohio River *E. caeruleum* group appears to have diverged from a common ancestor shared with the Lake Ontario–Upper Ohio River group described by Ray *et al*. (2006) by eight mutation steps and *c*. 1·12 mya [clades 2a and b; Fig. 4(b)].

Similar to the Lake Erie–Wabash River and Green River groups, the Lake Ontario–Upper Ohio River samples (clade 1b) show evidence of a rapid population expansion in the mismatch distribution analysis. The main stem Ohio River catchment group (clade 1a) is classified as a stationary or declining population in the mismatch distribution analysis. The mismatch distribution, however, is not significant when sampling sites from unglaciated areas are excluded, suggesting that Ohio River mainstream populations (Big Darby Creek and Little Miami River) rapidly expanded into the glaciated regions, as indicated by the star pattern in the network analysis [Fig. 4(b)].

DISCUSSION

BIOGEOGRAPHIC DIVERGENCE BETWEEN LAKE ERIE AND OHIO RIVER POPULATION GROUPS

Topographical alterations during the Pleistocene glaciations (Berendzen *et al*., 2003) destroyed older stream system connections and formed new ones, fragmenting

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fauna (Dyke & Prest, 1987; Pielou, 1991). At the end of the Pleistocene epoch (10 000–20 000 BP), the retreating Wisconsinan glaciers opened up waterways for migration of aquatic populations from southerly glacial refugia into newly created or reformed rivers (Bailey & Smith, 1981; Mandrak & Crossman, 1992; Lewis *et al*., 2008). Post-glacial dispersion pathways led to mixing of some fauna that had previously been genetically isolated, whereas others became geographically separated by different watersheds and then genetically diverged.

Notably, prior to the Ice Ages, the Teays River system flowed north-west and drained much of the Great Lakes region. It then was rerouted by the advancing ice to form the modern Ohio River system, which became isolated from the Great Lakes catchment and flowed south-west into the Mississippi River system (Mayden, 1988). Landscape genetic analyses presented here reveal that *E. caeruleum* populations in the Lake Erie and Ohio River catchments have been genetically separated since the early Pleistocene epoch *c*. 1·6 mya. Congruent with these results, Heithaus & Laushman (1997) recovered two *E. caeruleum* clades in the Lake Erie and Ohio River systems using allozyme data. In contrast, Ray *et al*. (2006) did not recognize the Lake Erie and Ohio River catchments as distinct groups using cyt *b* sequence data, likely due to their limited sampling and sole reliance on mtDNA.

As found in the present study, populations of the greenside darter *Etheostoma blennioides* Rafinesque, whose distribution is similar to *E. caeruleum*, diverge between the Ohio River and the Great Lakes–Wabash River catchments by *c*. 1·8 mya (Haponski & Stepien, 2008). Likewise, other Great Lakes fishes, including brook trout *Salvelinus fontinalis* (Mitchill) (Danzmann *et al*., 1998), walleye *Sander vitreus* (Mitchill) (Stepien & Faber, 1998; Stepien *et al*., 2009) and smallmouth bass *Micropterus dolomieu* Lacépède (Stepien *et al*., 2007), show this clear population genetic division between the watersheds. Separation between Ohio and Lake Erie clades of *S. vitreus* was dated at *c*. 1·5 mya (Stepien & Faber, 1998), congruent with the present results. Concordance of geographic and genetic patterns of the *E. caeruleum* with other Great Lakes species suggests that these taxa followed congruent recolonization pathways to the Great Lakes region from common glacial refugia.

LANDSCAPE GENETIC PATTERNS IN THE LAKE ERIE CATCHMENT

The west-to-east gradient of common *E. caeruleum* mtDNA haplotypes is a characteristic pattern from recolonization events following the end of the Pleistocene glaciations (Poissant *et al*., 2005). Greater genetic divergence of the Blanchard River sample in western Lake Erie from tributaries in the central basin (Chagrin, Cuyahoga and Grand Rivers) reflects their general geographic distances. The Blanchard River sample shares some mtDNA haplotypes with the other Lake Erie tributaries, but because the *E. caeruleum* has limited dispersal capabilities (Paine, 1984; Faber & White, 2001), this probably represents a common glacial refugium ancestry. The star pattern observed in the haplotype networks and mismatch distribution analyses suggests that haplotypes underwent rapid genetic radiation during the glacial–interglacial cycles. Closely spaced tributary populations in the central basin (Cuyahoga, Chagrin and Grand Rivers) are more genetically isolated from one another than would be expected according to an isolation by genetic distance hypothesis. Their appreciable population divergences reflect likely separation for thousands of years, experiencing

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little migration after initial post-Pleistocene colonization. *Micropterus dolomieu* populations in these three rivers likewise showed significant long-term divergence, with no recent gene flow (Stepien *et al*., 2007).

West to east haplotype gradients across Lake Erie have been observed for *E. blennoides* (Haponski & Stepien, 2008) and yellow perch *Perca flavescens* (Mitchill) (Sepulveda-Villet *et al*., 2009), with both having higher diversity in the west than in the central basin. *Sander vitreus* populations, however, show greatest genetic diversity in eastern Lake Erie (in samples located farther east than those in the present study), as well as a west-to-east gradient in the frequencies of specific genotypes (Stepien & Faber, 1998; Stepien *et al*., 2004; Strange & Stepien, 2007; Stepien *et al*., 2009). Higher diversity in eastern Lake Erie also characterizes *M. dolomieu* populations (Stepien *et al*., 2007). This higher diversity in the eastern basin has been attributed to population origins from an Atlantic glacial refugium shared with Lake Ontario (Stepien & Faber, 1998; Stepien *et al*., 2007, 2009), which needs to be further investigated for *E. caeruleum*. According to cyt *b* results, Lake Ontario samples of *E. caeruleum* and those from the upper Ohio River group together are more closely related to those from the Ohio River system, and markedly diverge from Lake Erie catchment populations. These data suggest that there were two colonization pathways for the lower Great Lakes system, respectively from the Mississippian and Atlantic coastal refugia (Mandrak & Crossman, 1992).

DNA results suggest that ancestry of western–central Lake Erie *E. caeruleum* genotypes traces to the Mississippian glacial refugium, indicated by their genetic relationship to those from the Wabash and Green Rivers (*c*. 1·09 mya). Lake Erie's formation dates to glacial Lake Maumee *c*. 14 000 BP, which then flowed west *via* the Ohio River to the Mississippi, changing its flow direction during several lake stages, to its current flow east to Lake Ontario *c*. 10 000 BP (Underhill, 1986). *Etheostoma caeruleum* colonizing newly formed Lake Erie probably travelled north-east from unglaciated portions of the Wabash River system, dispersing up through the Maumee River–glacial Lake Maumee outlet (Bailey & Smith, 1981; Mandrak & Crossman, 1992). This colonization pathway into the western Lake Erie catchment has been hypothesized for several other Great Lakes fishes, including lake whitefish (Bernatchez & Dodson, 1991), *M. dolomieu* (Stepien *et al*., 2007), *E. blennoides* (Haponski & Stepien, 2008), *P. flavescens* (Sepulveda-Villet *et al*., 2009) and *S. vitreus* (Stepien *et al*., 2009). Lake Erie tributaries were well connected by proglacial floodwaters and lakes (Bernatchez & Dodson, 1991) and dispersal into these new habitats probably occurred rapidly, resulting in founder effect and population bottlenecks, as suggested by *E. caeruleum* lower mtDNA diversity values and fewer haplotypes in comparison with main stem Ohio River populations.

LANDSCAPE GENETIC PATTERNS IN THE OHIO RIVER CATCHMENT

Congruent with their clinal pattern across Lake Erie, *E. caeruleum* populations in the Ohio River catchment show a west-to-east trend in frequencies of two common shared haplotypes; further sampling, however, is needed to determine if this pattern continues across a longer distance. Haplotypes in the main stem Ohio River system are more differentiated and exhibit population stationarity compared with those found in the Lake Erie–Wabash River catchments. Cytochrome *b* results suggest

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that ancestry of today's main stem Ohio River *E. caeruleum* populations traces to unglaciated portions of the Ohio River system in the Eastern Highlands (Fig. 1), which did not colonize the Great Lakes region. The haplotypes that recolonized the formerly glaciated portions of the Ohio River system probably expanded rapidly into them, as shown by the mismatch distribution analysis. Modern *E. caeruleum* populations in Lake Ontario and the upper Ohio River appear to trace their ancestry to the Atlantic slope refugium, where they diverged from main stem Ohio River types *c*. 1·12 mya.

Etheostoma caeruleum populations in the Ohio River sites at Big Darby Creek and Little Miami River have greater mtDNA genetic diversity and more private haplotypes than are found in Lake Erie samples. Congruent with these results, the allozyme study by Heithaus & Laushman (1997) recovered higher genetic diversity values in an Ohio River sampling site than found in the Lake Erie catchment, which they attributed to decreased water quality in the latter. However, this is probably a historic trend, with higher genetic diversity characteristic of populations from unglaciated regions than found in once glaciated areas (Billington & Hebert, 1991; Bernatchez & Wilson, 1998). Other Great Lakes fishes, including lake whitefish *Coregonus clupeaformis* (Mitchill) (Bernatchez & Dodson, 1991), *P. flavescens* (Todd & Hatcher, 1993; Sepulveda-Villet *et al*., 2009), black basses *Micropterus* spp. (Near *et al*., 2003; Stepien *et al*., 2007) and the *E. blennoides* (Haponski & Stepien, 2008) show this pattern of greater genetic diversity in unglaciated regions.

Etheostoma caeruleum probably recolonized the formerly glaciated portion of the Ohio River catchment (*i.e.* Big Darby Creek and Little Miami River) earlier than did the clade reaching the Lake Erie system, because the latter was under ice longer. Microsatellite genetic diversity levels, however, are similar between the two catchments. This difference reflects greater historical population bottlenecks in mtDNA, whose gene pool has approximately one-fourth the effective population size of nuclear DNA (Avise, 2004).

This study recovers two *E. caeruleum* population groups in the Lake Erie and Ohio River catchments, which diverged *c*. 1·6 mya during the early Pleistocene epoch. No mtDNA haplotypes are shared between the catchments and unique microsatellite alleles characterize each system today. Ohio River populations have higher mtDNA diversity values and more haplotypes than do populations in the Lake Erie catchment, attributed to their closer historic proximity to unglaciated source populations. *Etheostoma caeruleum* probably dispersed through the main stem of the Ohio River from unglaciated areas in the south and east. Today's Lake Ontario–Upper Ohio River population group traces to the Atlantic glacial refugium, where it diverged from the main stem Ohio River group *c*. 1·12 mya. Ancestors of the Lake Erie–Wabash River catchment group diverged *c*. 1·09 mya from the Green River group in the Mississippian glacial refugium, and then moved up the Wabash River to colonize early Lake Erie *c*. 14 000 BP. Fine-scale genetic divergence among Lake Erie tributary populations does not fit a geographic isolation by genetic distance pattern, probably due to low migration and genetic drift. Appreciable fine-scale genetic divergence among *E. caeruleum* populations in adjacent tributaries points to the need to conserve their habitats to maintain genetic diversity.

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